



MUTUAL FUND ANALYTICS REPORT

Bluestock Fintech Internship Program

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Executive Summary

The Mutual Fund Analytics Capstone Project was developed to analyze mutual fund performance, investor behavior, portfolio risk, and industry trends using a complete data analytics workflow. The project involved collecting, cleaning, transforming, and analyzing mutual fund datasets containing fund performance metrics, Net Asset Value (NAV) history, investor transactions, and fund scorecard information.

The project began with an ETL (Extract, Transform, Load) process where raw datasets were cleaned, validated, and structured for analysis. Exploratory Data Analysis (EDA) was then performed to identify patterns, trends, and relationships within the data. Key performance indicators such as Assets Under Management (AUM), returns, Sharpe ratio, alpha, drawdown, and investor transaction metrics were analyzed to evaluate mutual fund performance.

Advanced analytics techniques were applied to measure risk and investment quality. These included Value at Risk (VaR), rolling Sharpe ratio analysis, investor cohort analysis, SIP continuity analysis, and a rule-based fund recommendation model. These analyses provided deeper insights into portfolio risk, investor retention, and fund selection strategies.

To present the findings effectively, an interactive Power BI dashboard was developed consisting of multiple analytical pages. The dashboard enables users to explore industry trends, compare fund performance, analyze investor behavior, and monitor SIP and market-related insights through interactive visualizations and filters.

The final outcome of the project is a comprehensive mutual fund analytics solution that combines Python, SQL, and Power BI to support data-driven investment analysis and decision-making. The project demonstrates practical applications of data engineering, business intelligence, financial analytics, and dashboard development within the mutual fund domain.

Data Description

The Mutual Fund Analytics Project was developed using a comprehensive collection of mutual fund datasets covering fund performance, NAV history, investor transactions, SIP inflows, AUM trends, benchmark indices, and portfolio holdings. The objective of the dataset was to provide a complete view of the mutual fund ecosystem and support performance evaluation, risk analysis, investor behavior analysis, and dashboard development.

The project utilized ten primary raw datasets collected in CSV format. These datasets included fund master information, historical NAV data, Assets Under Management (AUM) records, monthly SIP inflows, category-wise investment flows, industry folio statistics, scheme performance metrics, investor transaction records, portfolio holdings, and benchmark index data.

Raw Datasets Used

Dataset	Description
Fund Master	Basic information about mutual fund schemes
NAV History	Historical Net Asset Value (NAV) records
AUM by Fund House	Assets Under Management across fund houses
Monthly SIP Inflows	Monthly SIP investment inflows
Category Inflows	Category-wise investment trends
Industry Folio Count	Industry-wide investor folio statistics
Scheme Performance	Returns and performance indicators
Investor Transactions	Investor investment and transaction records
Portfolio Holdings	Portfolio composition and holdings data
Benchmark Indices	Benchmark return and index information

To improve data quality and analytical usability, the raw datasets were processed using Python and Pandas. Various cleaning, transformation, and feature engineering techniques were applied to create structured analytical datasets.

Processed Datasets Generated

Several processed datasets were created during the project, including:

- clean_performance.csv
- clean_nav.csv
- clean_transactions.csv
- fund_scorecard.csv
- cagr_report.csv
- returns_computed.csv
- sharpe_values.csv
- sortino_values.csv
- max_drawdown.csv
- tracking_error.csv
- alpha_beta.csv

- cohort_analysis.csv
- sip_continuity.csv
- sector_hhi.csv
- var_cvar_report.csv

These processed datasets contain key investment and risk metrics such as CAGR, Alpha, Beta, Sharpe Ratio, Sortino Ratio, Maximum Drawdown, Tracking Error, Value at Risk (VaR), and Conditional Value at Risk (CVaR).

The final datasets were integrated into Power BI to create an interactive analytical dashboard. Relationships between datasets were established using common identifiers such as AMFI Code and Scheme Name, enabling comprehensive analysis across multiple dimensions including fund performance, risk, investor behavior, and industry trends.

The prepared data served as the foundation for exploratory data analysis, advanced financial analytics, dashboard development, and investment insights generated throughout the project.

ETL Process

Introduction

ETL (Extract, Transform, Load) is a critical step in the data analytics workflow. In this project, ETL was performed to collect data from multiple raw sources, clean and transform the data, generate analytical datasets, and prepare the information for dashboard development and advanced financial analysis.

Extract Phase

The extraction phase involved collecting mutual fund data from multiple CSV datasets containing information related to fund schemes, NAV history, AUM statistics, SIP inflows, investor transactions, portfolio holdings, benchmark indices, and industry-level metrics.

The extracted data served as the foundation for performance analysis, risk assessment, investor behavior analysis, and dashboard development.

Data was imported into Python using the Pandas library for further processing and validation.

Transform Phase

The transformation phase focused on improving data quality and preparing the datasets for analysis.

Data Cleaning

Several preprocessing operations were performed:

- Removal of duplicate records
- Handling of missing and null values
- Standardization of column names
- Correction of inconsistent data entries
- Conversion of date and numeric fields into appropriate formats
- Validation of fund identifiers and scheme information

Data Integration

Data from multiple sources was combined and linked using common identifiers such as AMFI Code and Scheme Name. This ensured consistency across different datasets and enabled cross-functional analysis.

Feature Engineering

Additional analytical metrics were generated using Python to support advanced financial analysis. These included:

- CAGR (Compound Annual Growth Rate)
- Alpha
- Beta
- Sharpe Ratio
- Sortino Ratio
- Maximum Drawdown
- Tracking Error
- Value at Risk (VaR)
- Conditional Value at Risk (CVaR)

Several intermediate and processed datasets were created during this stage, including fund scorecards, performance reports, cohort analysis reports, SIP continuity analysis, and risk measurement datasets.

Load Phase

After transformation, the cleaned and processed datasets were exported as structured CSV files and loaded into Power BI for visualization and dashboard development.

Relationships were established between datasets to create a unified analytical data model. These relationships enabled efficient filtering, aggregation, and cross-table analysis within the dashboards.

The final processed datasets were used to build interactive dashboards covering industry overview, fund performance analysis, investor analytics, SIP trends, and advanced risk metrics.

ETL Outcome

The ETL process successfully converted raw mutual fund data into a structured analytical model suitable for business intelligence and financial analytics. The resulting datasets provided a reliable foundation for exploratory data analysis, performance evaluation, risk assessment, and dashboard reporting throughout the project.

EDA Findings

Overview

Exploratory Data Analysis (EDA) was performed to understand the structure, quality, distribution, and trends present in the mutual fund datasets. The objective was to identify key patterns related to fund performance, investor behavior, SIP trends, Assets Under Management (AUM), and risk-return characteristics before conducting advanced analytics.

Data Quality Assessment

The raw datasets were examined for missing values, duplicate records, inconsistent formats, and invalid entries.

Key observations:

- Most datasets contained complete records with minimal missing values.
 - Duplicate records were identified and removed during preprocessing.
 - Column names and data formats were standardized across all datasets.
 - Numerical fields such as returns, NAV, AUM, and ratios were converted into appropriate numeric formats.
 - Date columns were transformed into datetime format for time-series analysis.
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Mutual Fund Performance Analysis

Analysis of fund performance datasets revealed significant variation across schemes and categories.

Key findings:

- Equity-oriented mutual funds generally delivered higher long-term returns.
 - Large-cap and flexi-cap funds showed relatively stable performance.
 - Small-cap and mid-cap funds generated higher returns but were associated with greater volatility.
 - Several schemes consistently outperformed their benchmark returns.
 - Sharpe Ratio analysis indicated that some funds generated superior risk-adjusted returns despite moderate absolute returns.
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Risk Analysis

Risk metrics were analyzed using Beta, Standard Deviation, Maximum Drawdown, Sharpe Ratio, and Sortino Ratio.

Key findings:

- Higher return funds generally exhibited higher risk levels.
 - Funds with high Sharpe Ratios demonstrated efficient risk-adjusted performance.
 - Maximum Drawdown values highlighted periods of significant downside risk.
 - Beta values showed varying sensitivity of schemes to market movements.
 - Risk and return displayed a positive correlation across most schemes.
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AUM Analysis

Assets Under Management (AUM) data was analyzed to understand fund size and industry concentration.

Key findings:

- A small number of fund houses accounted for a significant share of total industry AUM.
 - Larger funds generally attracted greater investor participation.
 - AUM growth showed a positive trend over the observed period.
 - Top-performing fund houses maintained consistently high asset bases.
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SIP Trend Analysis

Systematic Investment Plan (SIP) data was examined to evaluate retail investor participation.

Key findings:

- Monthly SIP inflows demonstrated an overall increasing trend.
 - Investor preference for systematic investing has grown steadily.
 - SIP contributions remained relatively stable even during periods of market volatility.
 - Long-term investment behavior was evident from sustained SIP participation.
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Investor Behaviour Analysis

Investor transaction records were analyzed to understand demographic and investment patterns.

Key findings:

- Investor participation varied across cities and regions.
 - Higher income groups generally invested larger amounts.
 - Growth plans were preferred over dividend-oriented schemes.
 - Long-term investors showed greater portfolio stability and retention.
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Benchmark Comparison

Fund returns were compared against benchmark indices.

Key findings:

- Several actively managed funds generated excess returns over benchmarks.
- Benchmark performance provided a useful reference for evaluating fund efficiency.
- Alpha analysis helped identify funds that consistently added value beyond market returns.

Performance Analysis

Overview

Performance analysis was conducted to evaluate mutual fund schemes based on return generation, risk exposure, risk-adjusted performance, and downside protection. Various financial performance indicators and advanced risk metrics were calculated to compare funds and identify high-quality investment options.

Return Analysis

Fund returns were evaluated using historical performance data and CAGR calculations.

Key observations:

- Long-term investment horizons generally produced better returns than short-term periods.
 - Equity-oriented schemes generated superior returns compared to conservative investment categories.
 - Certain schemes consistently outperformed both category averages and benchmark indices.
 - CAGR analysis provided a standardized measure of long-term growth performance across funds.
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Risk Analysis

Risk levels were assessed using Standard Deviation, Beta, and Maximum Drawdown metrics.

Key observations:

- Funds with higher returns generally exhibited greater volatility.
 - Standard Deviation values indicated varying levels of return fluctuations across schemes.
 - Beta values highlighted the sensitivity of funds to overall market movements.
 - Maximum Drawdown analysis identified schemes with significant downside risk during adverse market conditions.
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Sharpe Ratio Analysis

The Sharpe Ratio was used to evaluate risk-adjusted returns.

Key observations:

- Funds with higher Sharpe Ratios delivered better returns relative to the risk undertaken.
- Several schemes generated strong performance despite moderate volatility.
- Sharpe Ratio rankings helped identify efficient investment options among competing funds.

Sortino Ratio Analysis

The Sortino Ratio was calculated to measure returns relative to downside risk.

Key observations:

- Funds with high Sortino Ratios demonstrated effective downside protection.
- Negative return volatility was lower for top-performing schemes.
- Sortino analysis provided additional insight beyond traditional volatility measures.

Alpha and Beta Analysis

Alpha and Beta metrics were used to assess fund manager effectiveness and market sensitivity.

Key observations:

- Positive Alpha values indicated funds that generated returns above expected market performance.
- High Alpha funds demonstrated superior management efficiency.
- Beta values greater than one reflected higher market sensitivity.
- Beta values below one indicated relatively defensive investment behavior.

Maximum Drawdown Analysis

Maximum Drawdown was analyzed to evaluate potential losses during market downturns.

Key observations:

- Drawdown levels varied significantly across schemes.
- Funds with lower drawdowns demonstrated better capital preservation.
- High-return funds occasionally experienced deeper drawdowns due to increased market exposure.

Value at Risk (VaR) and Conditional Value at Risk (CVaR)

Advanced risk metrics were calculated to estimate potential losses under adverse market conditions.

Key observations:

- VaR estimated the maximum expected loss at a specified confidence level.
- CVaR measured the average loss beyond the VaR threshold.

- These metrics provided a more comprehensive assessment of tail-risk exposure.
 - Risk estimates supported portfolio risk management and investment decision-making.
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Fund Ranking and Scorecard Evaluation

A comprehensive scorecard was developed using multiple performance indicators.

Key observations:

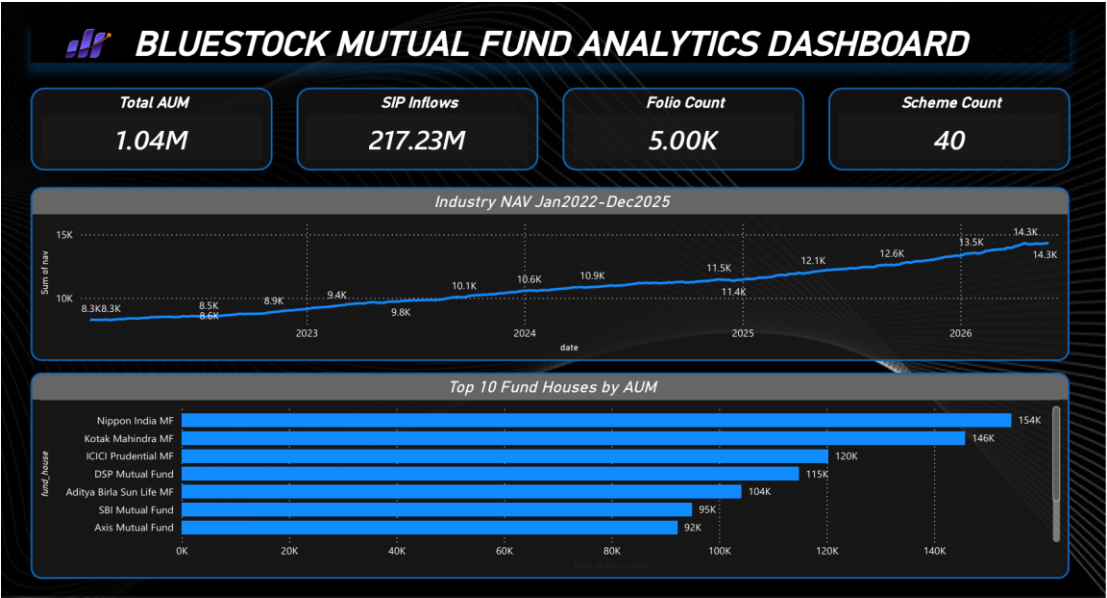
- Fund rankings incorporated return, risk, Sharpe Ratio, Alpha, and Drawdown metrics.
 - Top-ranked funds consistently demonstrated strong risk-adjusted performance.
 - Multi-factor evaluation reduced dependence on any single performance measure.
 - The scorecard provided a balanced framework for fund comparison and selection.
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Conclusion

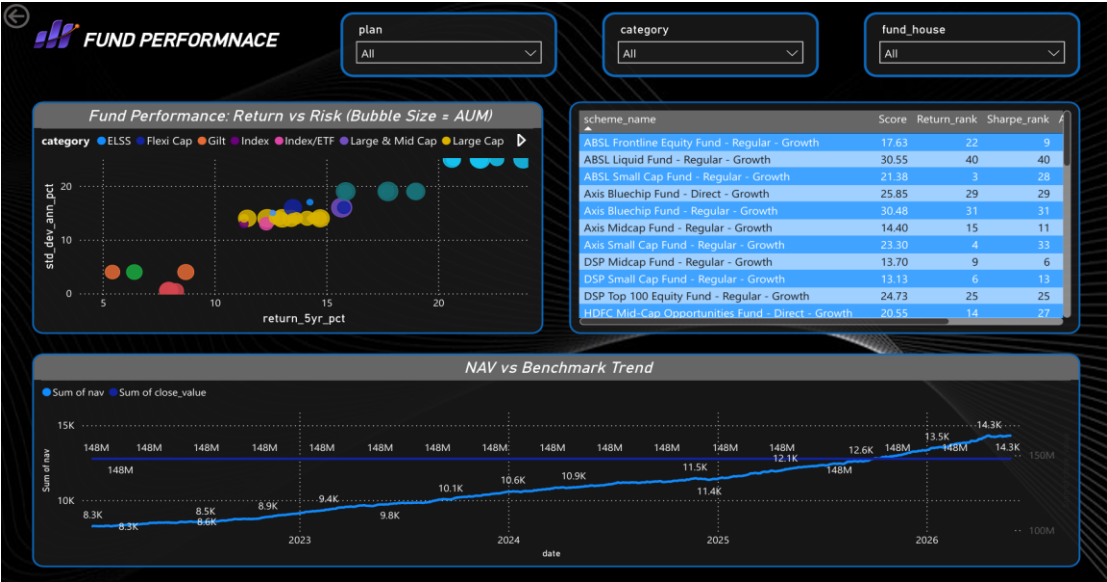
The performance analysis revealed that successful mutual fund selection requires balancing return generation with risk management. Metrics such as Sharpe Ratio, Sortino Ratio, Alpha, Drawdown, VaR, and CVaR provided deeper insights into fund quality beyond simple return comparisons. The combined analysis enabled identification of funds offering attractive risk-adjusted performance and long-term investment potential.

Dashboard Images

Industry Overview



Fund Performance



Investor Analysis



Recommendations

Based on the analysis performed throughout the project, the following recommendations are suggested:

1. Focus on Risk-Adjusted Performance

Investors should not rely solely on absolute returns while selecting mutual funds. Metrics such as Sharpe Ratio, Sortino Ratio, and Alpha should be considered to evaluate the quality and consistency of fund performance.

2. Encourage Long-Term SIP Investments

The SIP trend analysis indicates that systematic investing helps reduce the impact of market volatility. Investors should maintain long-term SIP contributions to benefit from rupee-cost averaging and compounding.

3. Diversify Across Fund Categories

Portfolio diversification across large-cap, mid-cap, flexi-cap, and sectoral funds can help balance risk and return while improving overall portfolio stability.

4. Monitor Downside Risk

Investors should evaluate Maximum Drawdown, VaR, and CVaR metrics before making investment decisions. Funds with controlled downside risk may provide better capital protection during market corrections.

5. Regular Benchmark Comparison

Fund performance should be periodically compared against relevant benchmark indices to identify underperforming schemes and make informed portfolio adjustments.

6. Utilize Data-Driven Decision Making

Financial institutions and investors can leverage analytical dashboards and performance scorecards to improve fund selection, portfolio monitoring, and investment planning.

7. Continuous Performance Tracking

Fund rankings and performance metrics should be updated regularly to reflect changing market conditions and ensure accurate investment evaluation.

Limitations

Despite providing valuable insights into mutual fund performance and investor behavior, this project has certain limitations:

1. Dependence on Historical Data

The analysis was performed using historical mutual fund data. Past performance may not accurately predict future market behavior or investment returns.

2. Limited Market Factors

External factors such as economic conditions, interest rate changes, inflation, government policies, and geopolitical events were not explicitly included in the analysis.

3. Data Availability Constraints

The accuracy of insights depends on the quality and completeness of the available datasets. Any missing or incomplete information may affect the results.

4. Simplified Risk Assumptions

Risk metrics such as Sharpe Ratio, Sortino Ratio, VaR, and CVaR are based on statistical assumptions and may not fully capture extreme market conditions.

5. Limited Investor Attributes

Investor behavior analysis was performed using the available demographic and transaction data. Additional behavioral and psychological factors were outside the scope of the dataset.

6. Dashboard Scope

The Power BI dashboard focuses on descriptive and analytical reporting. Advanced forecasting and real-time market integration were not implemented.

7. Static Dataset Environment

The project was developed using static datasets and does not automatically update with live market data.